

# CSA1280 – COMPUTER ARCHITECTURE

## EXPERIMENT – 17

### DIVISION OF TWO 16 – BIT NUMBERS

**AIM:** To write and execute an 8085 assembly language program to divide two 16 – bit numbers and store the result in memory.

#### ALGORITHM

1. Load the dividend into HL pair.
2. Load the divisor into DE pair.
3. Initialize the quotient to 0000H.
4. Compare the dividend with the divisor.
5. Subtract the divisor from the dividend.
6. Increment the quotient.
7. Repeat until the dividend is smaller than the divisor.
8. Store the quotient and remainder.
9. Stop the program.

#### CODE:

```
MOV AX,0020H
```

```
MOV BX,0004H
```

```
XOR DX,DX
```

```
DIV BX
```

```
MOV [8050H],AX
```

```
MOV [8052H],DX
```

```
HLT
```

## OUTPUT

